

Asymptotic Notation

Lecture 5

- Asymptotic Notations are mathematical tools used to analyze the performance of algorithms by understanding.
- These notations provide a concise way to express the behavior of an algorithm's time or space complexity as the input size approaches infinity.

Big-O Notation

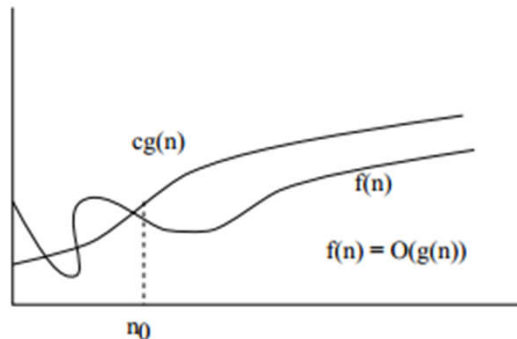
- **Definition:** $f(n) = O(g(n))$ iff there are two positive constants c and n_0 such that

$$|f(n)| \leq c |g(n)| \text{ for all } n \geq n_0$$

- If $f(n)$ is nonnegative, we can simplify the last condition to

$$0 \leq f(n) \leq c g(n) \text{ for all } n \geq n_0$$

- We say that “ $f(n)$ is big-O of $g(n)$.”
- As n increases, $f(n)$ grows no faster than $g(n)$. In other words, $g(n)$ is an *asymptotic upper bound* on $f(n)$.



Big-oh	Small-oh
If $f(n) = o(g(n))$ The bound, $0 \leq f(n) \leq c \cdot (g(n))$ holds for some constant $c > 0$ and all $n \geq n_0$	If $f(n) = o(g(n))$ The bound, $0 \leq f(n) < c \cdot (g(n))$ holds for all constant $c > 0$ and all $n > n_0$
Ex. $2n^2 = o(n^2)$ $\Rightarrow 2n^2 \leq c \cdot n^2$ for $c = 3$ above condition true.	Ex. $2n^2 = o(n^2)$ $2n^2 < c \cdot n^2$ for $c = 3, 4, \dots$ above condition is true but not for $c = 1$ hence, $2n^2 \neq o(n^2)$

Big- Ω notation

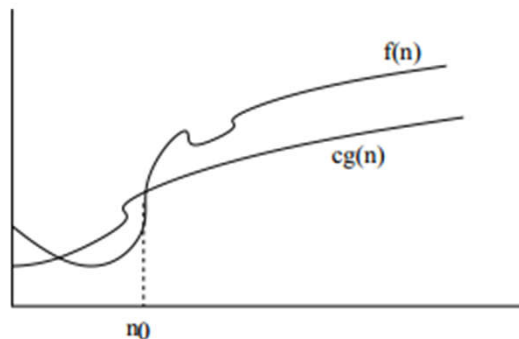
- **Definition:** $f(n) = \Omega(g(n))$ iff there are two positive constants c and n_0 such that

$$|f(n)| \geq c |g(n)| \text{ for all } n \geq n_0$$

- If $f(n)$ is nonnegative, we can simplify the last condition to

$$0 \leq c g(n) \leq f(n) \text{ for all } n \geq n_0$$

- We say that “ $f(n)$ is omega of $g(n)$.”
- As n increases, $f(n)$ grows no slower than $g(n)$. In other words, $g(n)$ is an *asymptotic lower bound* on $f(n)$.



Big-Omega (Ω)	Small-Omega (ω)
If $f(n) = \Omega(g(n))$ The bound, $0 \leq c \cdot (g(n)) \leq f(n)$ holds for some $c > 0$ and all $n \geq n_0$	If $f(n) = \omega(g(n))$ The bound, $0 \leq c \cdot (g(n)) < f(n)$ holds for all $c > 0$ and all $n \geq n_0$
Ex. $3n^2 = \Omega(n^2)$ $c \cdot n^2 < 3n^2$ for $c = 1$ above condition true Hence, $3n^2 = \Omega(n^2)$ is true.	Ex. $3n^2 = \omega(n^2)$ $\Rightarrow c \cdot n^2 < 3n^2$ for $c = 1, 2$ above condition is true but for $c \geq 3$, it becomes false hence, $3n^2 \neq \omega(n^2)$

Big- Θ notation

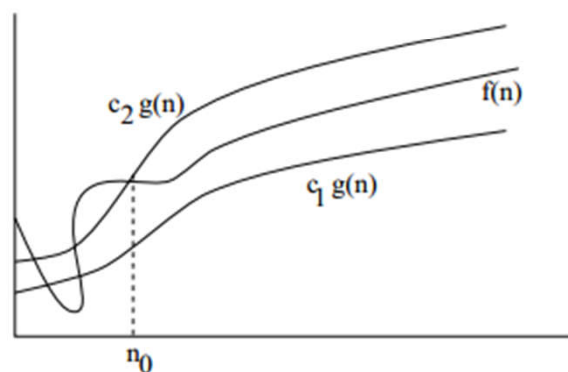
- **Definition:** $f(n) = \Theta(g(n))$ iff there are three positive constants c_1 , c_2 and n_0 such that

$$c_1|g(n)| \leq |f(n)| \leq c_2|g(n)| \text{ for all } n \geq n_0$$

- If $f(n)$ is nonnegative, we can simplify the last condition to

$$0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for all } n \geq n_0$$

- We say that “ $f(n)$ is theta of $g(n)$.”
- As n increases, $f(n)$ grows at the same rate as $g(n)$. In other words, $g(n)$ is an *asymptotically tight bound* on $f(n)$.



1.5 Comparisons of Functions

Many of the relational properties apply to asymptotic comparisons as well i.e. functions can be compared asymptotically.

1. Transitivity:

$$f(n) = \Theta(g(n)) \text{ and } g(n) = \Theta(h(n)) \text{ imply } f(n) = \Theta(h(n))$$

$$f(n) = O(g(n)) \text{ and } g(n) = O(h(n)) \text{ imply } f(n) = O(h(n))$$

$$f(n) = \Omega(g(n)) \text{ and } g(n) = \Omega(h(n)) \text{ imply } f(n) = \Omega(h(n))$$

$$f(n) = o(g(n)) \text{ and } g(n) = o(h(n)) \text{ imply } f(n) = o(h(n))$$

$$f(n) = \omega(g(n)) \text{ and } g(n) = \omega(h(n)) \text{ imply } f(n) = \omega(h(n))$$

2. Reflexivity:

$$f(n) = \Theta(f(n)),$$

$$f(n) = O(f(n)),$$

$$f(n) = \Omega(f(n)).$$

3. Symmetry:

$$f(n) = \Theta(g(n)) \text{ if and only if } g(n) = \Theta(f(n)).$$

4. Transpose Symmetry:

$$f(n) = O(g(n)) \text{ if and only if } g(n) = \Omega(f(n)),$$

$$f(n) = o(g(n)) \text{ if and only if } g(n) = \omega(f(n)).$$

5. Because these properties hold for asymptotic notations, one can draw an analogy between the asymptotic comparison of two function f and g and the comparison of two real number a and b :

$$f(n) = O(g(n)) \approx a \leq b,$$

$$f(n) = \Omega(g(n)) \approx a \geq b,$$

$$f(n) = \Theta(g(n)) \approx a = b,$$

$$f(n) = O(g(n)) \approx a < b,$$

$$f(n) = \omega(g(n)) \approx a > b.$$